

6.03	Algebraic equations				
6.03a	Linear equations in one unknown	Solve linear equations in one unknown algebraically. e.g. Solve $3x - 1 = 5$	Set up and solve linear equations in mathematical and non-mathematical contexts, including those with the unknown on both sides of the equation. e.g. Solve $5(x - 1) = 4 - x$ Interpret solutions in context.	<i>[Examples may include manipulation of algebraic fractions, 6.01g]</i>	A3, A17, A21
6.03b	Quadratic equations		Solve quadratic equations with coefficient of x^2 equal to 1 by factorising. e.g. Solve $x^2 - 5x + 6 = 0$. Find x for an x cm by $(x + 3)$ cm rectangle of area 40cm^2 .	Know the quadratic formula. Rearrange and solve quadratic equations by factorising, completing the square or using the quadratic formula. e.g. $2x^2 = 3x + 5$ $\frac{2}{x} - \frac{2}{x+1} = 1$	A18

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6.03c	Simultaneous equations		Set up and solve two linear simultaneous equations in two variables algebraically. e.g. Solve simultaneously $2x + 3y = 18$ and $y = 3x - 5$	Set up and solve two simultaneous equations (one linear and one quadratic) in two variables algebraically e.g. Solve simultaneously $x^2 + y^2 = 50$ and $2y = x +$
6.03d	Approximate solutions using a graph	Use a graph to find the approximate solution of a linear equation.	Use graphs to find approximate roots of quadratic equations and the approximate solution of two linear simultaneous equations.	Know that the coordinates of the points of intersection of a curve and a straight line are the solutions to the simultaneous equations for the line and curve.
6.03e	Approximate solutions by iteration			Find approximate solutions to equations using systematic sign-change methods (for example, decimal search or interval bisection) when there is no simple analytical method of solving them. Specific methods will not be requested in the assessment.

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